Sunil Karichilamuttath Velayudhan

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QUALIFICATION PROFILE

AI/ML professional with a Master's in Artificial Intelligence and expertise in data science, machine learning, and cloud infrastructure. Experienced in developing AI solutions in NLP, computer vision, and robotics, with a strong background in deep learning, big data analytics, and cloud architectures. Proven ability to drive innovation, manage complex projects, and collaborate effectively across teams to deliver impactful solutions in diverse industries.

EDUCATION

MSc	Artificial Intelligence Royal Holloway University of London	**2023**
BSc	Computer Science and Engineering Cochin University of Science and Technology, Kerala - india	**2017**
Diploma	Computer Engineering Govt. Polytechnic College Purappuzha, Kerala, India	**2014**

TECHNICAL EXPERTISE

Data Analysis | Machine Learning | Computer Vision | Deep Learning | Natural Language Processing Large Language Models |Artificial Neural Networks | Convolutional neural Networks | Artificial Intelligence | Statistical Data Analysis |Big Data Analytics |PCA | PyTorch Keras| Autonomous Systems | MLOps | Data Cleaning | Python | R | MATLAB | OpenCV | VGG | ResNet | Inception(GoogleNet) EfficientNet | DenseNet | YOLO | Faster R-CNN | DeepLab | FCN | scikit-image | Pillow(PL) | COCO Roboflow | Java | IT Infrastructure | Azure | AWS Devops | AWS – SageMaker | Spark | Power BI JupyterNotebook | Google Colab | Visual studio code | Anaconda | Kaggle discord

PROFESSIONAL EXPERIENCE

Data Analyst **Aug 2023 - May 2025**
Axiom Technologies Pvt.ltd- London, United Kingdom
Client : BBC Studios London, Standard Chartered

- Developed and maintained dashboards and reports to support data-driven decision-making for BBC Studios and Standard Chartered.
- Performed data extraction, cleaning, and transformation using SQL, Python, and Excel.

- Analyzed key performance indicators (KPIs) to identify trends, opportunities, and risks.
- Collaborated with cross-functional teams to gather requirements and deliver actionable insights.
- Automated repetitive data processes, reducing manual effort and increasing efficiency by 30%.
- Presented findings and recommendations to stakeholders using data visualization tools such as Power BI and Tableau.

Data Analyst **Sep 2021 - Aug 2023**
Scholar Co ltd- Manchester, United Kingdom

- Collected, cleaned, and analyzed large datasets to identify trends and provide actionable insights for business decision-making.
- Developed and maintained dashboards and reports, using tools such as Power BI and Excel, to visualize data and communicate findings to stakeholders.
- Collaborated with cross-functional teams to assess data needs, ensuring the provision of accurate and timely data for analysis.
- Conducted statistical analyses to identify patterns, correlations, and business opportunities, improving operational efficiency and service delivery.
- Automated data collection and reporting processes, enhancing workflow efficiency and ensuring data accuracy.
- Provided ad-hoc analytical support for ongoing projects and assisted in the interpretation of results to inform strategic decisions.

Software Engineer **Apr 2021 - Aug 2021**
Overbrook Technology Services Private Limited
Kochi, Kerala, India

- Reviewed and analyzed the business requirement document and functional documents by working as Scrum QA.
- Utilized selenium with JAVA to understand framework and application for creating automation test cases.
- Responsible for automating inline scrum stories and presenting scrum stories demo.
- Managed and created test data that was used in test design and execution.
- Researched and developed test scripts along with debugging by expertly using Selenium.
- Gather relevant information and prepare daily and weekly status reported to stakeholders.
- Prepared automation handover document by collecting relevant information and data.
- Designed SQL queries to validate data and count records for DB tables.

Additional experience as Engineer II in Java Development at Spotfixcrew and Junior Software Engineer at Woft Technologies Pvt.ltd

PROJECTS

Regression models of Gender differences in Automotive purchase : Developed regression models to analyze gender differences in automotive purchases using different regression models). The project aimed to predict purchasing patterns and identify key factors influencing gender-specific preferences in the automotive market. Dataset creation using Beautiful Soup library|LinearRegression|Lasso, Ridge Regression, and Support Vector Machines (SVM) |Python - Jupyter notebook

Breast Cancer Classification with Deep Learning : Developed and evaluated advanced deep learning models (VGG16 and Xception, both with attention mechanisms) for breast cancer image classification. Processed medical imaging data into three diagnostic categories using data augmentation and attention layers to enhance model focus. Achieved ~89% accuracy, with Xception outperforming in precision and recall for complex patterns. Demonstrated expertise in medical imaging, model evaluation (precision, recall, F1-score), and advanced neural network architectures.

Customer Segmentation Using Clustering : Developed a customer segmentation model using clustering techniques such as K-Means, DBSCAN, and PCA. The project aimed to identify distinct customer groups based on purchasing behavior and demographics, optimizing marketing strategies and im- proving customer targeting through data-driven insights. It uses libraries such as pandas, numpy, matplotlib, seaborn, and vari- ous scikit-learn components - StandardScaler| (PCA)|KMeans | DBSCAN

Home Price Prediction in California : Developed a machine learning model to predict home prices in California using a dataset of housing attributes such as location, median income, and proximity to the ocean. Applied various algorithms, including Linear Regression, Random Forest, and XGBoost, with hyperparameter tuning via Grid Search. Evaluated model performance using Mean Squared Error (MSE) and R- squared (R2). Technologies used include Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

WORKSHOPS AND TRAININGS

- **Advanced Medical Imaging Workshop: Hands-On Classification & Object Detection with Deep Learning:**
- **Medical Imaging Workshop:** Deep Learning with YOLO
- **Generative AI Workshop by Google & Kaggle**
- **Mastering LLM-Optimization using Langchain Workshop**
  Attended a workshop on optimizing LLMs with Langchain, focusing on Retrieval-Augmented Generation (RAG) architecture and enhancing model performance for NLP tasks.

LANGUAGE

English (professional), Hindi (professional), Malayalam (Native), Tamil (Native)